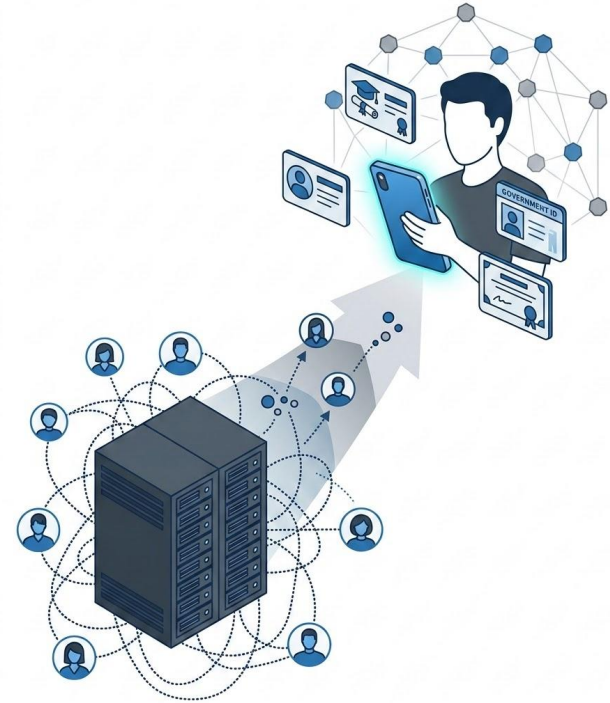


Blockchain and Cryptocurrencies Technologies

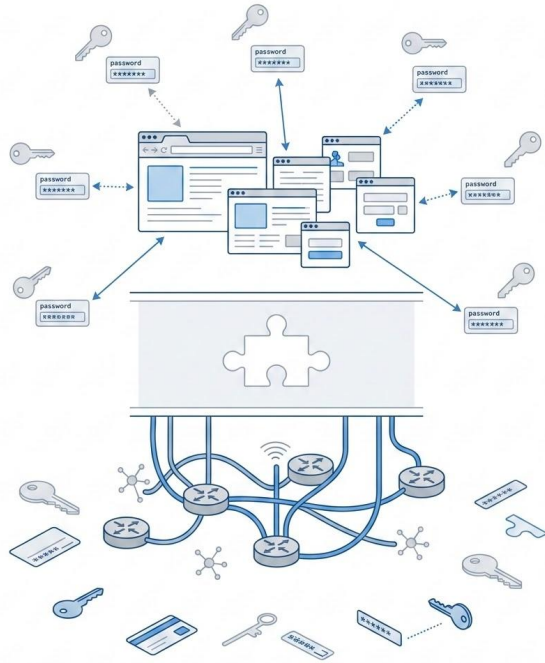
Self Sovereign Identity and Blockchain Trust
From Centralized Accounts to Verifiable Claims

Self Sovereign Identity and Blockchain Trust

- From centralized accounts to verifiable claims
- Blockchain and Cryptocurrencies Technologies
- Digital identity as a trust infrastructure problem
- User controlled credentials as an alternative to platform controlled accounts
- SSI shows both the power and the limits of blockchain based trust



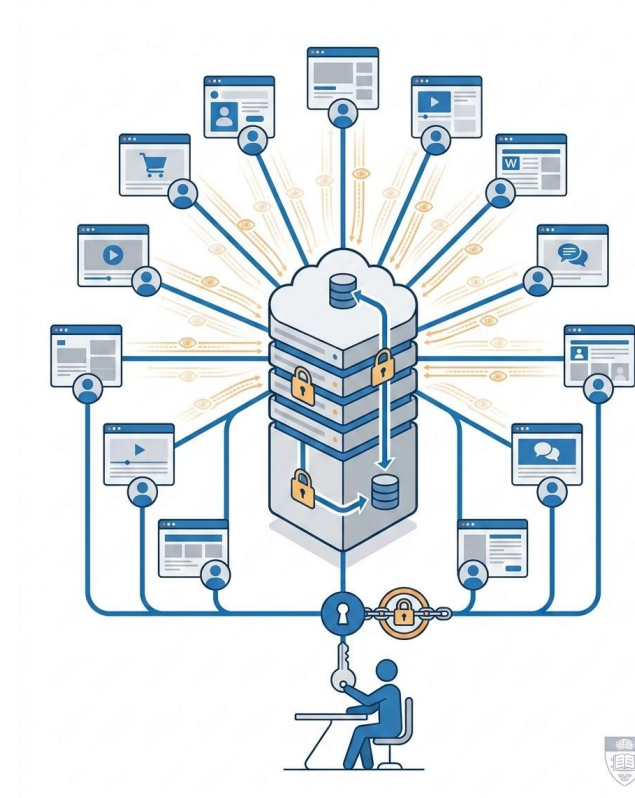
The Internet Was Built Without Identity



- TCP/IP solved communication, not identity
- Identity became the missing layer of the Internet stack
- Web systems added authentication later as an application layer patch
- Web 1.0 created scattered usernames and passwords
- The result was weak authentication, poor usability, and fragmented trust
- Blockchain enters the story because it offers public verification without a single operator

Web 2.0 Solved Identity by Centralizing Power

- Web 2.0 made login convenient by concentrating identity in large providers
- Google, Meta, Apple and others became identity providers
- OAuth and federated login improved usability
- Authentication events became behavioral data
- Centralized identity creates surveillance, lock in, bans, and breach concentration
- SSI reacts against identity platforms becoming identity monopolies



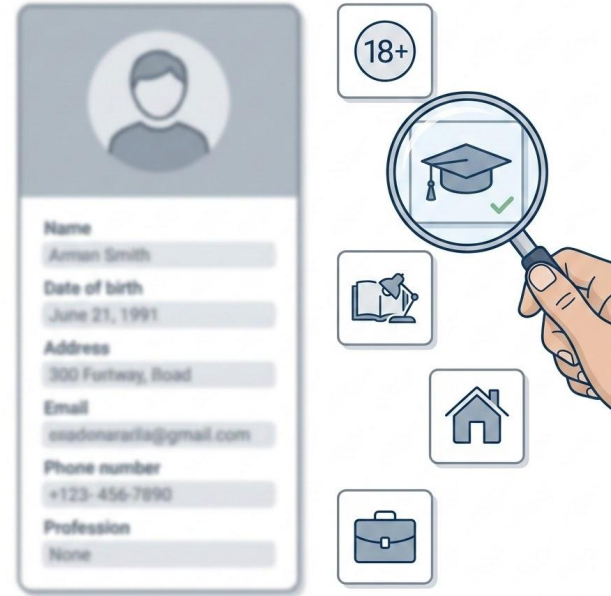
The Blockchain Question for Identity



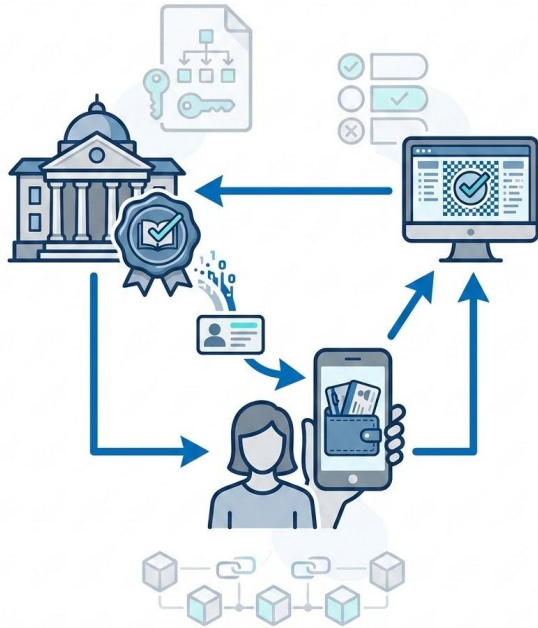
- Can trust be verified without a central provider?
- Can identity claims be verified without depending on one login provider?
- Can users hold credentials like they hold private keys?
- Can verifiers check authenticity without calling the issuer every time?
- Can revocation and trust status be public, auditable, and tamper resistant?
- SSI tries to answer these questions with cryptography, wallets, and sometimes ledgers

From Accounts to Claims

- Identity becomes something you prove, not only something you log into
- Traditional systems ask: “Who are you?”
- SSI systems ask: “What can you prove?”
- Identity becomes a set of verifiable claims
- Examples include being enrolled, being over 18, holding a degree, being a professor, or owning a rural property
- This matches blockchain thinking: verify proofs, not institutional promises



The SSI Trust Triangle

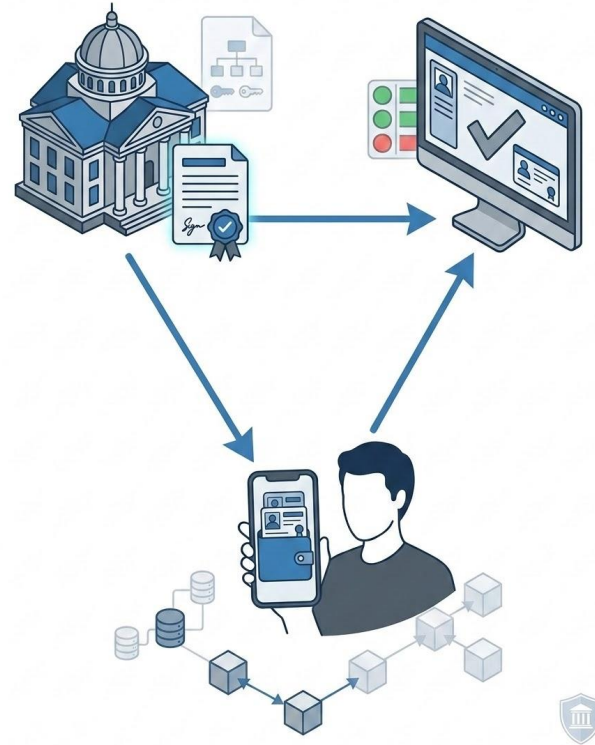


- SSI is built around Issuer, Holder, and Verifier
- Issuer signs claims and credentials
- Holder stores credentials and controls presentations
- Verifier checks authenticity, status, and issuer authority
- This replaces direct database lookup with cryptographic presentation
- Trust does not disappear. It moves to issuer governance, credential schemas, and verification infrastructure

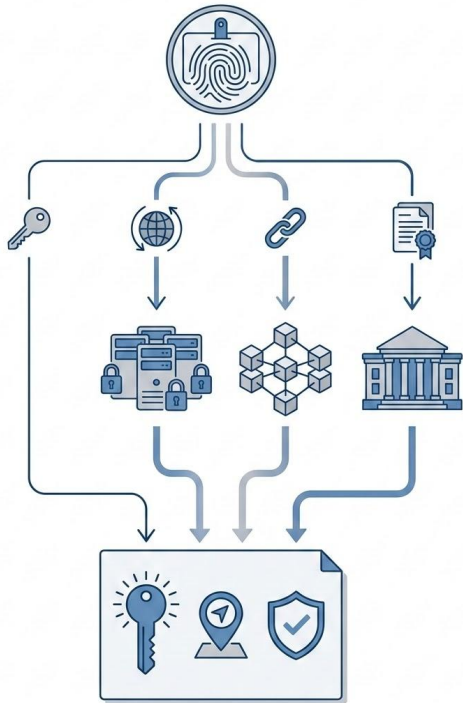


Identity Credentials vs Attribute Credentials

- Not everything in SSI is “identity”
- Personal identity credentials identify the person
- Examples include PID, national ID, and passport
- Attribute credentials qualify the person
- Examples include student status, professor status, degree, professional license, and course competence
- SSI becomes powerful when identity is decomposed into reusable attributes
- Blockchain relevance: attributes can be verified independently and composed into proofs



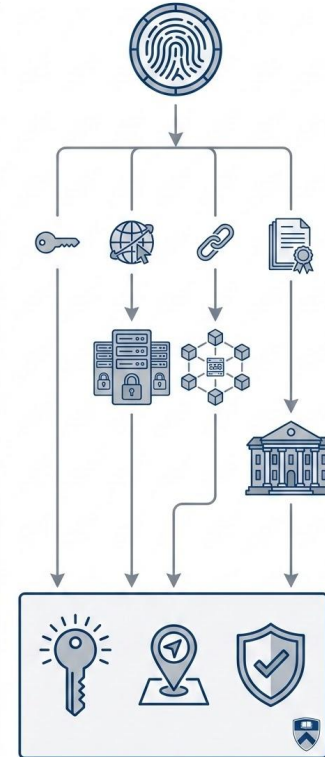
Verifiable Credentials



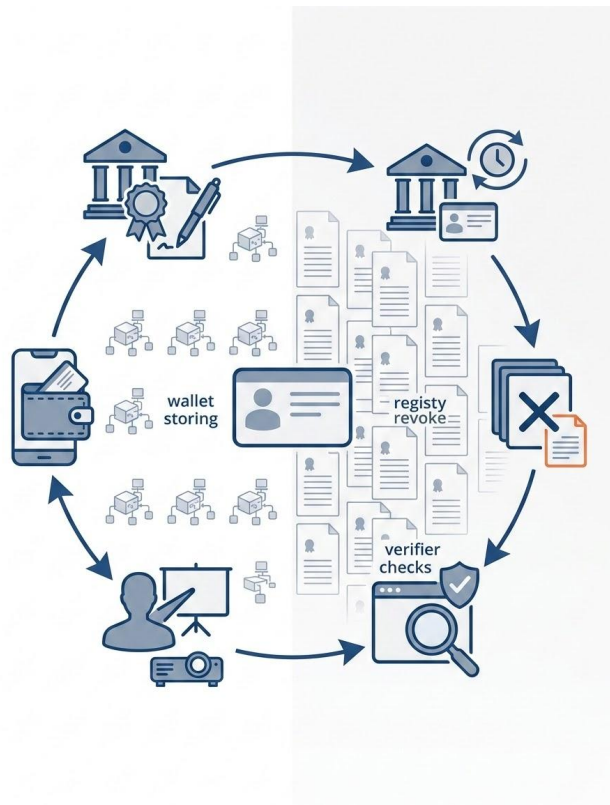
- Verifiable credentials are structured claims, not just digital PDFs
- A verifiable credential is a signed, portable, machine verifiable claim
- It is issued by a trusted issuer and held by the user
- The verifier checks issuer signature, credential schema, credential status, and issuer authority
- VCs are digital documents designed for protocol level verification
- They enable selective disclosure and automated trust decisions

Decentralized Identifiers

- DIDs are identifiers with resolvable trust metadata
- A DID identifies a subject without requiring one global identity provider
- DID methods define how identifiers are created and resolved
- DID resolution returns a DID Document with public keys, service endpoints, and verification methods
- DID methods may depend on keys, web domains, ledgers, or registries
- Important point: DIDs do not automatically eliminate centralized trust



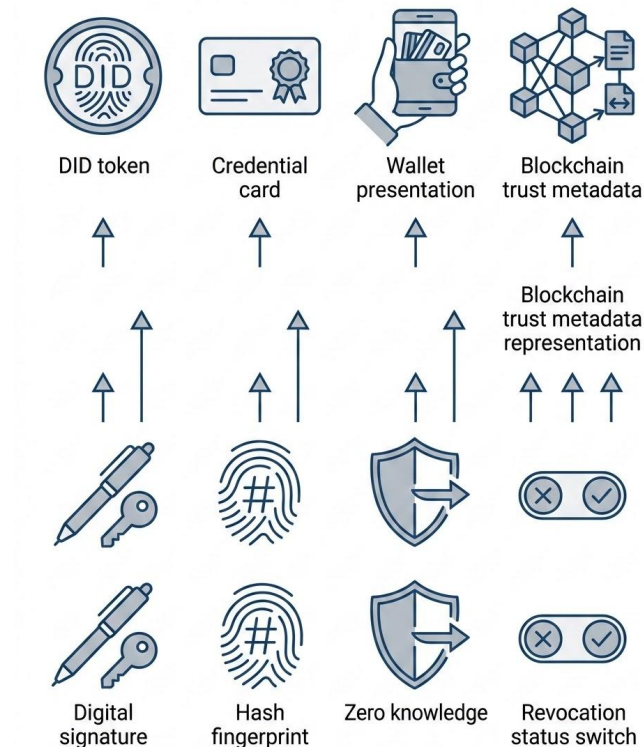
Credential Lifecycle



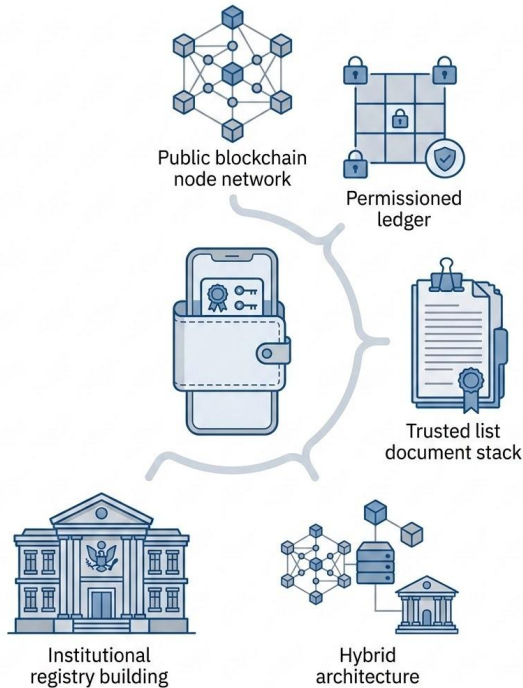
- Trust changes over time
- Credentials are not static objects
- A complete lifecycle includes issue, store, present, verify, revoke, renew, and reissue
- Blockchain can help with timestamping, revocation status, DID resolution, and auditability
- Lifecycle governance remains institutional
- A valid signature is not enough if the credential is expired, revoked, or unauthorized

Cryptographic Foundations Revisited

- This is why the primitives lecture matters
- Digital signatures allow issuers to sign credentials and holders to prove control
- Hashes provide integrity and commitments
- Zero knowledge proofs enable predicate proofs without raw data disclosure
- Revocation registries provide dynamic trust status
- SSI is an application layer built on the same cryptographic foundations as blockchain
- Image prompt



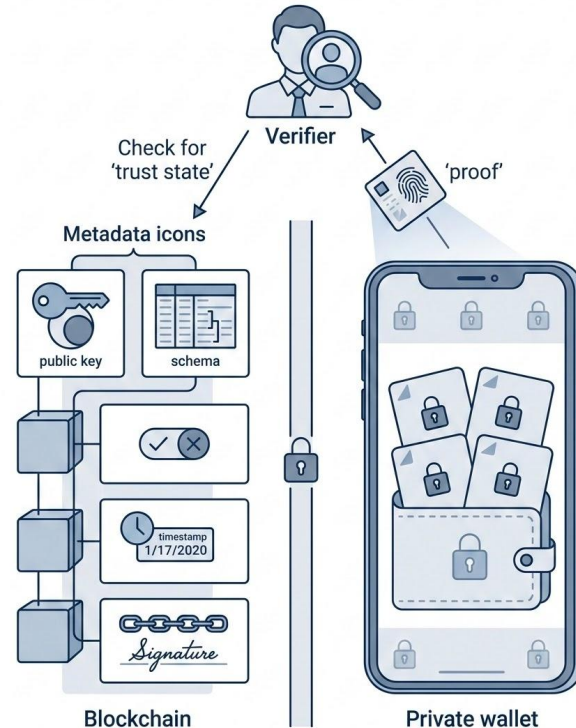
Blockchain Is Optional, Not Mandatory



- Architecture depends on the trust problem
- SSI can be implemented with public blockchains, permissioned ledgers, trusted lists, institutional registries, or hybrid models
- Blockchain is useful for shared public state, tamper resistance, censorship resistance, and availability
- Blockchain is not ideal for simple governance, legal accountability, low complexity, or easy interoperability
- The question is not “can we use blockchain?”
- The question is “what trust problem are we solving?”

What Should Go On Chain?

- Public verification belongs on chain, personal data does not
- Do not put personal data on chain
- Good candidates for on chain data include DID metadata, issuer public keys, revocation registries, credential schema references, timestamps, and commitments
- Personal data should remain off chain in the holder wallet
- The blockchain should verify trust state, not store identity dossiers
- Rule: public verification on chain, private attributes off chain



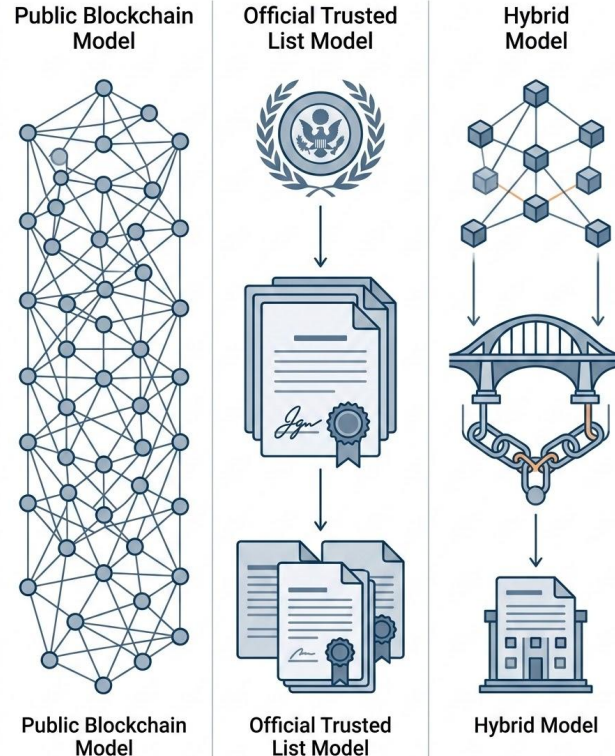
Hybrid SSI Architecture



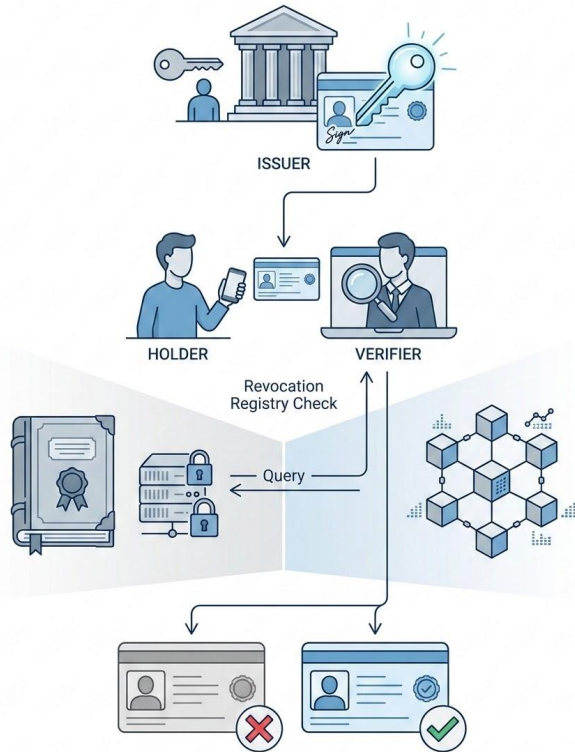
- Hybrid SSI combines wallets, ledgers, trusted lists, and institutions
- Holder wallet stores credentials and private keys
- Issuer publishes verification metadata
- Ledger or trusted list provides public trust state
- Verifier checks presentation proof, issuer authorization, revocation status, and schema compliance
- Hybrid trust combines user control, public verification, and institutional accountability

Blockchain Based Trust vs Trusted Lists

- Two trust models, different tradeoffs
- Public blockchain model: trust root is ledger consensus, strength is availability and tamper resistance, weakness is governance and interoperability
- Trusted list model: trust root is legal authority, strength is accountability and cross border recognition, weakness is centralization
- Hybrid model: practical balance with higher complexity
- Real identity systems usually need governance, not only decentralization



Revocation Is the Hard Reality Check



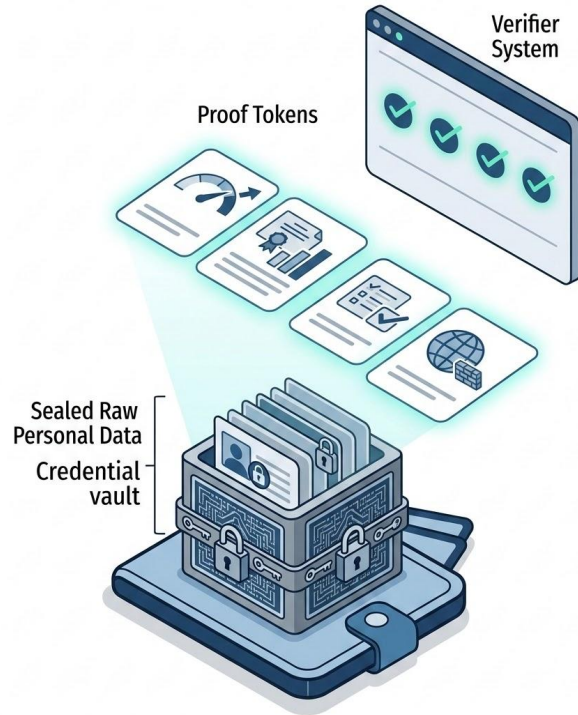
- A valid signature is not enough
- Credentials may need revocation because of fraud, expiration, loss of status, issuer error, or compromised keys
- Traditional revocation is centralized, slow, and hard to enforce
- SSI can use cryptographic revocation registries
- Blockchain can make revocation public, timestamped, and auditable
- Revocation may also create privacy and correlation risks

Selective Disclosure

- Selective disclosure changes identity from document presentation to claim presentation
- Traditional documents over disclose
- SSI allows disclosure of only required attributes
- Examples include proving nationality without full ID, enrollment without course or semester, degree completion without grades, and professor status without unrelated identity data
- Selective disclosure makes privacy part of the protocol design



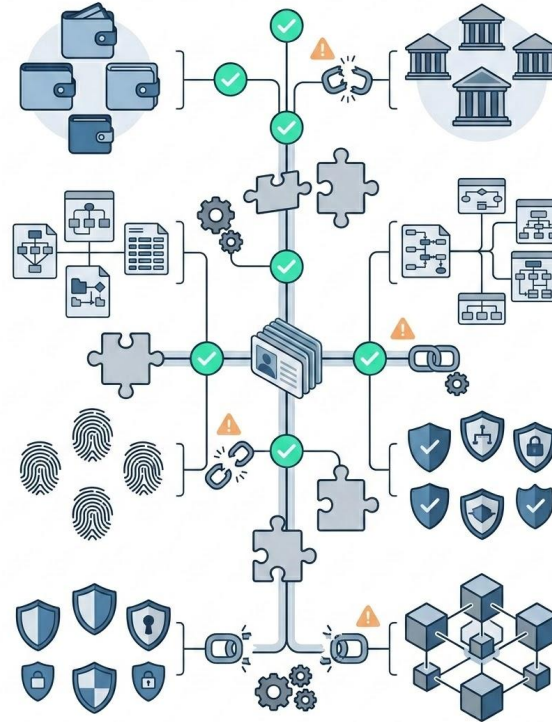
Zero Knowledge Proofs in SSI



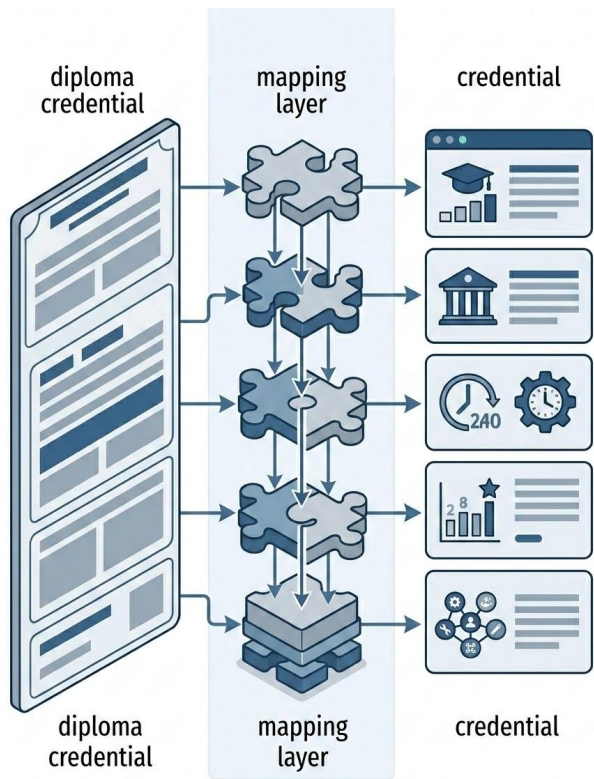
- ZK proofs allow predicates to be proven without revealing raw data
- Examples include age greater than 18, IAA greater than 9.5, valid KYC completed, and not being on a sanctions list
- ZK connects SSI with blockchain privacy technologies
- ZK does not solve governance: someone defines the claim, someone trusts the issuer, and someone validates the proof
- Privacy needs cryptography and institutional meaning

Interoperability Is the Real Battlefield

- The hardest problem is not always cryptography
- SSI fails if credentials cannot move across wallets, countries, schemas, DID methods, trust frameworks, and blockchains
- Technical interoperability requires common syntax
- Semantic interoperability requires common meaning
- Legal interoperability requires mutual recognition
- Blockchain can worsen fragmentation if every ecosystem anchors trust differently



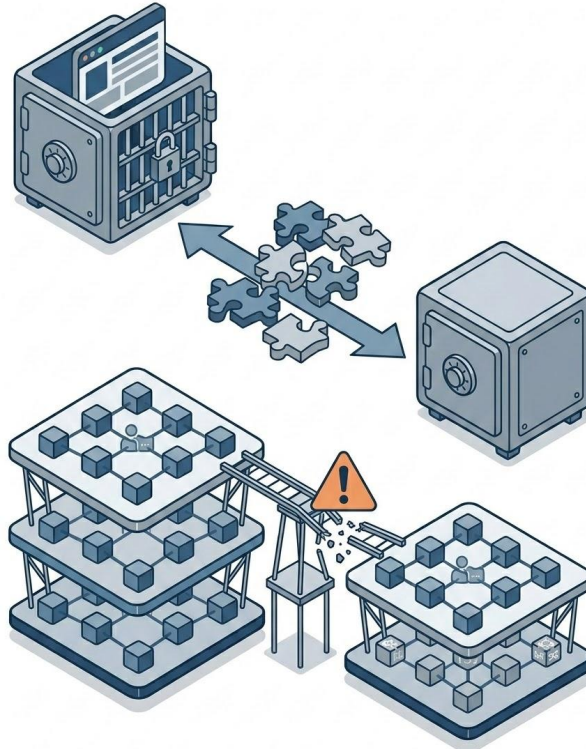
Schemas, Languages, and Meaning



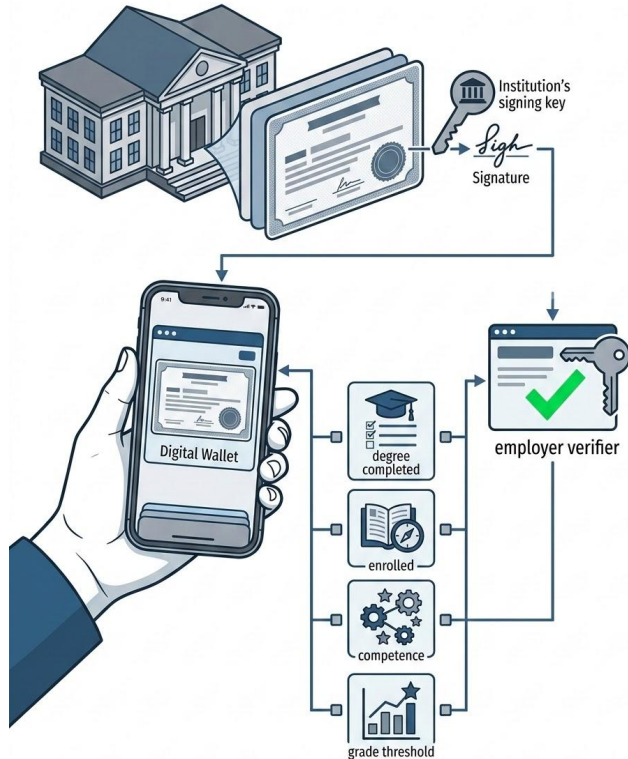
- A digital diploma is not interoperable just because it is digital
- Fields must map across jurisdictions: course, degree level, institution, workload, grades, and competencies
- Language matters: Portuguese tags, Spanish tags, and European schemas may encode the same meaning differently
- The goal is not to invent isolated national credentials
- The goal is to adapt local documents to global credential schemas
- Interoperability requires shared meaning, not only shared technology

Wallet Lock In and Cross Chain Trust Bridges

- Portability is the promise, lock in is the risk
- Wallets may transform credentials internally
- Export and import can fail across wallets
- DID methods may depend on different infrastructures
- Blockchains introduce cross chain trust problems
- Bridges are complex and can become security bottlenecks
- Standardization by W3C, DIF, ISO, ETSI and others is still evolving
- The system is only sovereign if the user can actually move



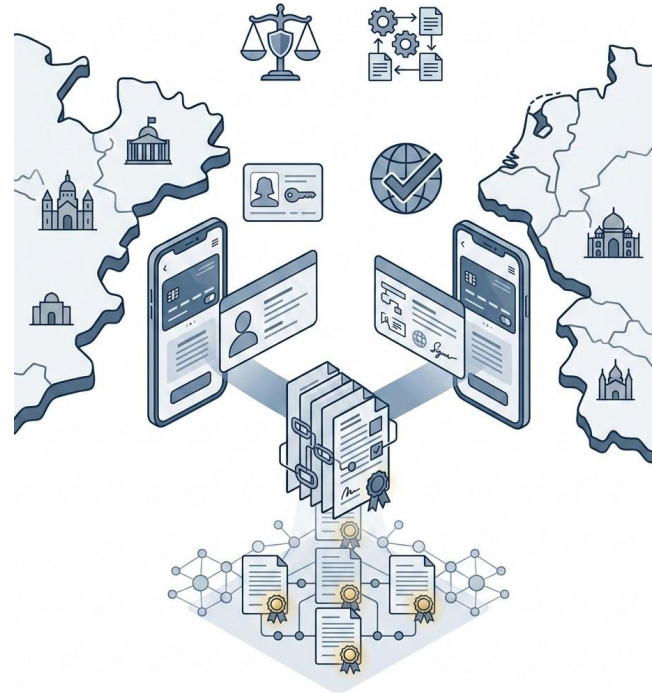
Case Study: Digital Diplomas as Verifiable Credentials



- Brazilian digital documents can connect with global VC models
- Digital diplomas are a strong SSI use case
- Universities issue degree credentials
- Students store them in wallets
- Employers verify authenticity without notarization or PDF checking
- Selective disclosure enables proving degree completion, enrollment, competence, or grade threshold
- This connects Brazilian digital documents with global verifiable credential models

Case Study: eIDAS2, EUDI Wallet, and Brazil

- Trust frameworks shape SSI adoption
- eIDAS2 moves Europe toward regulated digital identity wallets
- The EU model relies strongly on trusted lists and legal recognition
- It supports cross border identity and credential verification
- Brazil is likely to align through ICP Brasil as a trust framework, gov.br initiatives, CIN digital evolution, and Mercosur and EU digital market integration
- SSI adoption will be shaped as much by law as by blockchain



Financial KYC, Healthcare, Ethics, and Discussion



- SSI must be designed for context, not ideology
- Financial sector: reusable KYC proofs, lower onboarding cost, less document custody, FATF traceability pressure, and possible ZK KYC
- Healthcare: prescriptions, vaccination, insurance, and highly sensitive credentials
- Critical tension: selective disclosure protects privacy, but partial information can create risk in clinical decisions
- Final question: can SSI be user controlled and still regulated?

Final Takeaways

- SSI is not just identity on blockchain
- It is a trust architecture for proving claims with cryptographic assurance
- Blockchain can help with public verification, timestamping, revocation, auditability, and availability
- Trusted lists can help with legal recognition, accountability, and cross border governance
- The hard problems are not only cryptographic: they are interoperability, recovery, usability, governance, and regulation
- The real challenge is building user controlled identity without rebuilding centralized surveillance with better cryptography

